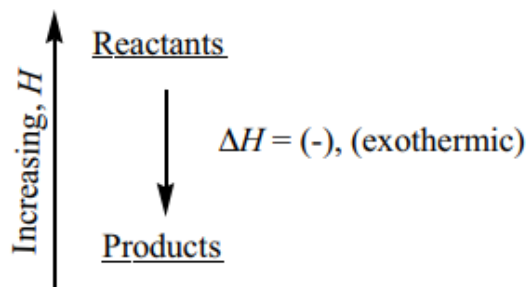
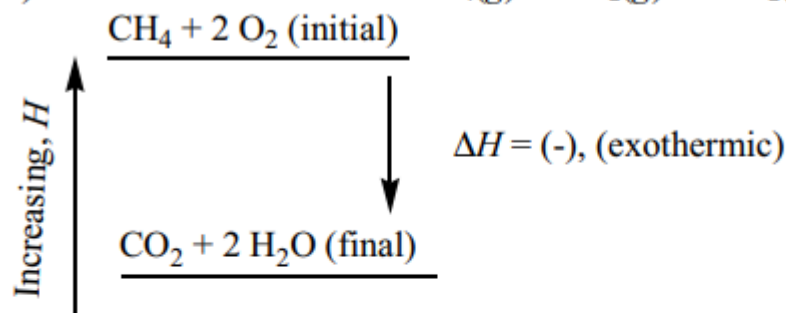


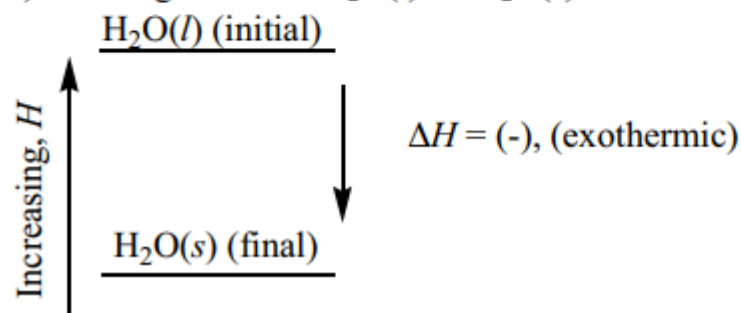
- 6.6 The amount of the change in internal energy in the two cases is the same. By the law of energy conservation, the change in energy of the universe is zero. This requires that the change in energy of the system (heater or air conditioner) equals an opposite change in energy of the surroundings (room air). Since both systems consume the same amount of electrical energy, the change in energy of the heater equals that of the air conditioner.
- 6.8 The change in a system's energy is $\Delta E = q + w$. If the system receives heat, then its q_{final} is greater than q_{initial} so q is positive. Since the system performs work, its $w_{\text{final}} < w_{\text{initial}}$ so w is negative. The change in energy is $(+425 \text{ J}) + (-425 \text{ J}) = \mathbf{0 \text{ J}}$.
- 6.10 A system that releases thermal energy has a negative value for q and a system that has work done on it has a positive value for work. So,
 $\Delta E = -675 \text{ J} + (525 \text{ cal} \times 4.184 \text{ J/cal}) = 1521.6 = \mathbf{1.52 \times 10^3 \text{ J}}$
- 6.12 $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 3.3 \times 10^{10} \text{ J}$
 (1.0 ton)
- a) $\Delta E(\text{kJ}) = (3.3 \times 10^{10} \text{ J}) \left(\frac{1 \text{ kJ}}{10^3 \text{ J}} \right) = \mathbf{3.3 \times 10^7 \text{ kJ}}$
- b) $\Delta E(\text{kcal}) = (3.3 \times 10^{10} \text{ J}) \left(\frac{1 \text{ cal}}{4.184 \text{ J}} \right) \left(\frac{1 \text{ kcal}}{10^3 \text{ cal}} \right) = 7.887 \times 10^6 = \mathbf{7.9 \times 10^6 \text{ kcal}}$
- c) $\Delta E(\text{Btu}) = (3.3 \times 10^{10} \text{ J}) \left(\frac{1 \text{ Btu}}{1055 \text{ J}} \right) = 3.12796 \times 10^7 = \mathbf{3.1 \times 10^7 \text{ Btu}}$
- 6.15 $\text{Time} = (1.0 \text{ lb}) \left(\frac{4.1 \times 10^3 \text{ Cal}}{1.0 \text{ lb}} \right) \left(\frac{10^3 \text{ cal}}{1 \text{ Cal}} \right) \left(\frac{4.184 \text{ J}}{1 \text{ cal}} \right) \left(\frac{1 \text{ kJ}}{10^3 \text{ J}} \right) \left(\frac{\text{h}}{1850 \text{ kJ}} \right) = 9.2726 = \mathbf{9.3 \text{ hour}}$
- 6.17 Since many reactions are performed in an open flask, the reaction proceeds at constant pressure. The determination of ΔH (constant pressure conditions) requires a measurement of heat only, whereas ΔE requires measurement of heat and PV work.
- 6.19 a) **Exothermic**, the system (water) is releasing heat in changing from liquid to solid.
 b) **Endothermic**, the system (water) is absorbing heat in changing from liquid to gas.
 c) **Exothermic**, the process of digestion breaks down food and releases energy.
 d) **Exothermic**, heat is released as a person runs and muscles perform work.
 e) **Endothermic**, heat is absorbed as food calories are converted to body tissue.
 f) **Endothermic**, the wood being chopped absorbs heat (and work).
 g) **Exothermic**, the furnace releases heat from fuel combustion. Alternatively, if the system is defined as the air in the house, the change is endothermic since the air's temperature is increasing by the input of heat energy from the furnace.
- 6.22 An exothermic reaction releases heat, so the reactants have greater H (H_{initial}) than the products (H_{final}).
 $\Delta H = H_{\text{final}} - H_{\text{initial}} < 0$.



- 6.24 a) Combustion of methane: $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g}) + \text{heat}$



- b) Freezing of water: $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{s}) + \text{heat}$



- 6.33 Plan: The heat required to raise the temperature of water by $80.^{\circ}\text{C}$ is found by using equation 6.7, or $q = c \times \text{mass} \times \Delta T$. The specific heat capacity, c_{water} , is found in Table 6.4. Because the Celsius degree is the same size as the kelvin degree, $\Delta T = 80.^{\circ}\text{C} = 80. \text{ K}$.

$$q(\text{J}) = (\text{mass}) C (\Delta T) = (12.0 \text{ g}) \left(4.184 \frac{\text{J}}{\text{g}^{\circ}\text{C}} \right) ((100. - 20.)^{\circ}\text{C}) = 4016.64 = 4.0 \times 10^3 \text{ J}$$

- 6.35 $q(\text{J}) = (\text{mass}) C (\Delta T)$
 $85.0 \text{ kJ}(10^3 \text{ J/kJ}) = (295 \text{ g}) (0.900 \text{ J/g}^{\circ}\text{C}) (T_f - 3.00)^{\circ}\text{C}$

$$(T_f - 3.00)^{\circ}\text{C} = \frac{(85.0 \text{ kJ}) \left(\frac{10^3 \text{ J}}{1 \text{ kJ}} \right)}{(295 \text{ g}) \left(\frac{0.900 \text{ J}}{\text{g}^{\circ}\text{C}} \right)} = 320.15^{\circ}\text{C} \text{ (unrounded)}$$

$$T_f = \mathbf{323^{\circ}\text{C}}$$

- 6.37 Since the bolts have the same mass, and one must cool as the other heats, the intuitive answer is $[(T_1 + T_2) / 2] = \mathbf{77.5^{\circ}\text{C}}$.

- 6.39 Use the same procedure in 6.37, but mass is now a factor because the two water samples are not identical. Both volumes are converted to mass using the density (e.g., $85 \text{ mL} \times 1.00 \text{ g/mL} = 85 \text{ g}$). Convert $^{\circ}\text{C}$ temperatures (22°C , 82°C) to Kelvin temperatures (295 K , 355 K).

$$-q_{\text{lost}} = q_{\text{gained}}$$

$$- [85 \text{ mL} (1.00 \text{ g/mL})] (4.184 \text{ J/g}^{\circ}\text{C}) (T_f - 82)^{\circ}\text{C} = [165 \text{ mL} (1.00 \text{ g/mL})] (4.184 \text{ J/g}^{\circ}\text{C}) (T_f - 22)^{\circ}\text{C}$$

$$- [85](T_f - 82) = [165] (T_f - 22)$$

$$6970 - 85 T_f = 165 T_f - 3630$$

$$6970 + 3630 = 165 T_f + 85 T_f$$

$$10600. = 250. T_f$$

$$T_f = (10600. / 250.) = 42.4 = \mathbf{42^{\circ}\text{C}}$$

- 6.41 Plan: Heat gained by water plus heat lost by copper tubing must equal zero, so $q_{\text{water}} = -q_{\text{copper}}$. However, some heat will be lost to the insulated container.

Solution:

$$-q_{\text{lost}} = q_{\text{gained}} = q_{\text{water}} + q_{\text{calorimeter}}$$

$$-(505 \text{ g Cu})(0.387 \text{ J/g}^\circ\text{C})(T_f - 99.9)^\circ\text{C} = (59.8 \text{ g H}_2\text{O})(4.184 \text{ J/g}^\circ\text{C})(T_f - 24.8)^\circ\text{C} + (10.0 \text{ J/}^\circ\text{C})(T_f - 24.8)^\circ\text{C}$$

$$-(195.435)(T_f - 99.9) = (250.2032)(T_f - 24.8) + (10.0)(T_f - 24.8)$$

$$19523.9565 - 195.435 T_f = 250.2032 T_f - 6205.03936 + 10.0 T_f - 248$$

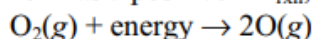
$$19523.9565 + 6205.03936 + 248 = 250.2032 T_f + 195.435 T_f + 10.0 T_f$$

$$19523.9565 + 6205.03936 + 248 = 250.2032 T_f + 195.435 T_f + 10.0 T_f$$

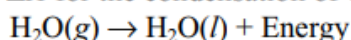
$$25976.99586 = (455.6382) T_f$$

$$T_f = 25976.99586 / (455.6382) = 57.0123 = \mathbf{57.0^\circ\text{C}}$$

- 6.48 The reaction has a positive ΔH_{rxn} , because this reaction requires the input of energy to break the O - O bond:



- 6.49 As a substance changes from the gaseous state to the liquid state, energy is released so ΔH would be negative for the condensation of 1 mol of water. The value of ΔH for the vaporization of 2 mol of water would be twice the value of ΔH for the condensation of 1 mol of water vapor but would have an opposite sign ($+\Delta H$).



$$\Delta H_{\text{cond}} = (-)$$

It would be opposite in sign to and one-half the value for the vaporization of 2 moles of liquid H_2O to H_2O vapor.

- 6.50 a) This reaction is **exothermic** because ΔH is negative.
 b) Because ΔH is a state function, the total energy required for the reverse reaction, regardless of how the change occurs, is the same magnitude but different sign of the forward reaction. Therefore, $\Delta H = +20.2 \text{ kJ}$.
 c) The ΔH_{rxn} is specific for the reaction as written, meaning that 20.2 kJ is released when 1/8 of a mole of sulfur reacts. In this case, 3.2 moles of sulfur react and we therefore expect that much more energy will be released.

$$\Delta H_{\text{rxn}} = (3.2 \text{ mol S}_8) \left(\frac{-20.2 \text{ kJ}}{(1/8) \text{ mol S}_8} \right) = -517.12 = \mathbf{-5.2 \times 10^2 \text{ kJ}}$$

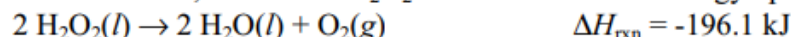
- d) The mass of S_8 requires conversion to moles and then a calculation identical to c) can be performed.

$$\Delta H_{\text{rxn}} = (20.0 \text{ g S}_8) \left(\frac{1 \text{ mol S}_8}{256.56 \text{ g S}_8} \right) \left(\frac{-20.2 \text{ kJ}}{(1/8) \text{ mol S}_8} \right) = -12.5974 = \mathbf{-12.6 \text{ kJ}}$$

- 6.52 a) $1/2 \text{ N}_2(\text{g}) + 1/2 \text{ O}_2(\text{g}) \rightarrow \text{NO}(\text{g}) \quad \Delta H = 90.29 \text{ kJ}$

$$\text{b) } \Delta H_{\text{rxn}} = (1.50 \text{ g NO}) \left(\frac{1 \text{ mol NO}}{30.01 \text{ g NO}} \right) \left(\frac{-90.29 \text{ kJ}}{1 \text{ mol NO}} \right) = -4.5129957 = \mathbf{-4.51 \text{ kJ}}$$

- 6.54 For the reaction written, 2 moles of H_2O_2 release 196.1 kJ of energy upon decomposition.

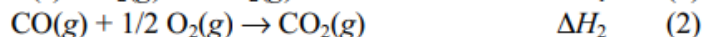


$$\text{Heat} = q = (732 \text{ kg H}_2\text{O}_2) \left(\frac{10^3 \text{ g}}{1 \text{ kg}} \right) \left(\frac{1 \text{ mol H}_2\text{O}_2}{34.02 \text{ g H}_2\text{O}_2} \right) \left(\frac{-196.1 \text{ kJ}}{2 \text{ mol H}_2\text{O}_2} \right) = -2.1097 \times 10^6 = \mathbf{-2.11 \times 10^6 \text{ kJ}}$$

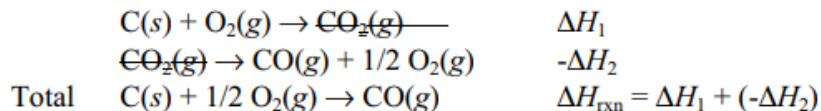
- 6.58 a) $\text{C}_2\text{H}_4(\text{g}) + 3 \text{ O}_2(\text{g}) \rightarrow 2 \text{ CO}_2(\text{g}) + 2 \text{ H}_2\text{O}(\text{l}) \quad \Delta H_{\text{rxn}} = -1411 \text{ kJ}$

$$\text{b) Mass C}_2\text{H}_4 = (-70.0 \text{ kJ}) \left(\frac{1 \text{ mol C}_2\text{H}_4}{-1411 \text{ kJ}} \right) \left(\frac{28.04 \text{ g C}_2\text{H}_4}{1 \text{ mol C}_2\text{H}_4} \right) = 1.39107 = \mathbf{1.39 \text{ g C}_2\text{H}_4}$$

- 6.62 Two chemical equations can be written based on the description given:

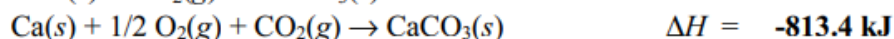


The second reaction can be reversed, its ΔH sign changed, and appropriate coefficients can be multiplied to one or both of the equations to allow addition of the two reactions. In this case, no coefficients are necessary because the CO_2 cancels.



How are the ΔH 's for each reaction determined? The reverse of equation (2) was used in Sample Problem 6.7, with $\Delta H_2 = -283.0 \text{ kJ}$. The ΔH_1 can be found by using the heats of formation in Appendix B: $\Delta H_1 = \Delta H_f(\text{CO}_2) = -393.5 \text{ kJ/mol} - (0 + 0) = -393.5 \text{ kJ/mol}$. Assume one mole for equation (1) and $\Delta H_{\text{rxn}} = \Delta H_1 - \Delta H_2 = -393.5 \text{ kJ} - (-283.0 \text{ kJ}) = \mathbf{-110.5 \text{ kJ}}$.

- 6.63 To obtain the overall reaction, add the first reaction to the reverse of the second.



- 6.65 Vaporization is the change in state from a liquid to a gas. The two equations describing these chemical reactions can be combined to yield the equation for vaporization.



- 6.67 Equation 3 is: $\text{N}_2\text{(g)} + 2 \text{O}_2\text{(g)} \rightarrow 2 \text{NO}_2\text{(g)} \quad \Delta H_{\text{rxn}} = +66.4 \text{ kJ}$

In Figure P6.67, A represents reaction 1 with a larger amount of energy absorbed, B represents reaction 2 with a smaller amount of energy released and C represents reaction 3 as the sum of A and B.

- 6.70 The standard heat of reaction, $\Delta H_{\text{rxn}}^\circ$, is the enthalpy change for any reaction where all substances are in their standard states. The standard heat of formation, ΔH_f° , is the enthalpy change that accompanies the formation of one mole of a compound in its standard state from elements in their standard states. Standard state is 1 atm for gases, 1 M for solutes, and pure state for liquids and solids. Standard state does not include a specific temperature, but a temperature must be specified in a table of standard values.

- 6.72 a) $1/2 \text{Cl}_2\text{(g)} + \text{Na(s)} \rightarrow \text{NaCl(s)}$ The element chlorine occurs as Cl_2 , not Cl.
 b) $\text{H}_2\text{(g)} + 1/2 \text{O}_2\text{(g)} \rightarrow \text{H}_2\text{O(l)}$ The element hydrogen exists as H_2 , not H, and the formation of water is written with water as the product in the liquid state.
 c) No changes

- 6.73 a) $\text{Ca(s)} + \text{Cl}_2\text{(g)} \rightarrow \text{CaCl}_2\text{(s)}$
 b) $\text{Na(s)} + 1/2 \text{H}_2\text{(g)} + \text{C(graphite)} + 3/2 \text{O}_2\text{(g)} \rightarrow \text{NaHCO}_3\text{(s)}$
 c) $\text{C(graphite)} + 2 \text{Cl}_2\text{(g)} \rightarrow \text{CCl}_4\text{(l)}$
 d) $1/2 \text{H}_2\text{(g)} + 1/2 \text{N}_2\text{(g)} + 3/2 \text{O}_2\text{(g)} \rightarrow \text{HNO}_3\text{(l)}$

- 6.75 The enthalpy change of a reaction is the sum of the ΔH_f° of the products minus the sum of the ΔH_f° of the reactants. Since the ΔH_f° values (Appendix B) are reported as energy per one mole, use the appropriate coefficient to reflect the higher number of moles.

$$\Delta H_{\text{rxn}}^\circ = \sum [\Delta H_f^\circ (\text{products})] - \sum [\Delta H_f^\circ (\text{reactants})]$$

$$\begin{aligned} \text{a) } \Delta H_{\text{rxn}}^\circ &= 2 \Delta H_f^\circ [\text{SO}_2\text{(g)}] + 2 \Delta H_f^\circ [\text{H}_2\text{O(g)}] - 2 \Delta H_f^\circ [\text{H}_2\text{S(g)}] - 3 \Delta H_f^\circ [\text{O}_2\text{(g)}] \\ &= 2 \text{ mol}(-296.8 \text{ kJ/mol}) + 2 \text{ mol}(-241.826 \text{ kJ/mol}) - 2 \text{ mol}(-20.2 \text{ kJ/mol}) - 3(0.0) \\ &= \mathbf{-1036.8 \text{ kJ}} \end{aligned}$$

$$\begin{aligned} \text{b) The balanced equation is } &\text{CH}_4\text{(g)} + 4 \text{Cl}_2\text{(g)} \rightarrow \text{CCl}_4\text{(l)} + 4 \text{HCl(g)} \\ \Delta H_{\text{rxn}}^\circ &= 1 \text{ mol}(-139 \text{ kJ/mol}) + 4 \text{ mol}(-92.31 \text{ kJ/mol}) - 1 \text{ mol}(-74.87 \text{ kJ/mol}) - 4 \text{ mol}(0) \\ &= \mathbf{-433 \text{ kJ}} \end{aligned}$$

6.77 Use Hess's Law to solve for heat of formation of copper(II) oxide.



$$\Delta H_{\text{rxn}}^{\circ} = 2 \text{ mol } (\Delta H_f^{\circ}, \text{CuO}(s)) - 1 \text{ mol } (\Delta H_f^{\circ}, \text{Cu}_2\text{O}(s)) - 1/2 \text{ mol } (\Delta H_f^{\circ}, \text{O}_2(g))$$

$$-146.0 \text{ kJ} = 2 \text{ mol } (\Delta H_f^{\circ}, \text{CuO}(s)) - 1 \text{ mol } (-168.6 \frac{\text{kJ}}{\text{mol}}) - 1/2 \text{ mol } (0)$$

$$-146.0 \text{ kJ} = 2 \text{ mol } (\Delta H_f^{\circ}, \text{CuO}(s)) + 168.6 \text{ kJ}$$

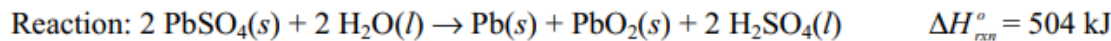
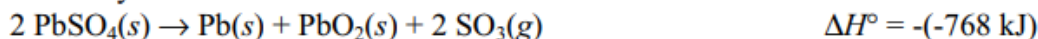
$$\Delta H_f^{\circ}, \text{CuO}(s) = -\frac{314.6 \text{ kJ}}{2 \text{ mol}} = \mathbf{-157.3 \text{ kJ/mol}}$$

6.80 $2 \text{PbSO}_4(s) + 2 \text{H}_2\text{O}(l) \rightarrow \text{Pb}(s) + \text{PbO}_2(s) + 2 \text{H}_2\text{SO}_4(l)$

$$\text{a) } \Delta H_{\text{rxn}}^{\circ} = 1 \text{ mol } (\Delta H_f^{\circ}, \text{Pb}(s)) + 1 \text{ mol } (\Delta H_f^{\circ}, \text{PbO}_2(s)) + 2 \text{ mol } (\Delta H_f^{\circ}, \text{H}_2\text{SO}_4(l)) - 2 \text{ mol } (\Delta H_f^{\circ}, \text{PbSO}_4(s)) \\ - 2 \text{ mol } (\Delta H_f^{\circ}, \text{H}_2\text{O}(l))$$

$$= 1 \text{ mol } (0 \frac{\text{kJ}}{\text{mol}}) + 1 \text{ mol } (-276.6 \frac{\text{kJ}}{\text{mol}}) + 2 \text{ mol } (-813.989 \frac{\text{kJ}}{\text{mol}}) - 2 \text{ mol } (-918.39 \frac{\text{kJ}}{\text{mol}}) - 2 \text{ mol } (-285.840 \frac{\text{kJ}}{\text{mol}}) \\ = \mathbf{503.9 \text{ kJ}}$$

b) Reverse the first equation (changing the sign of $\Delta H_{\text{rxn}}^{\circ}$) and multiply the coefficients (and $\Delta H_{\text{rxn}}^{\circ}$) of the second reaction by 2.



6.81 a) $\text{C}_{18}\text{H}_{36}\text{O}_2(s) + 26 \text{O}_2(g) \rightarrow 18 \text{CO}_2(g) + 18 \text{H}_2\text{O}(g)$

$$\text{b) } \Delta H_{\text{comb}}^{\circ} = 18 \text{ mol } (\Delta H_f^{\circ}, \text{CO}_2(g)) + 18 \text{ mol } (\Delta H_f^{\circ}, \text{H}_2\text{O}(g)) - 1 \text{ mol } (\Delta H_f^{\circ}, \text{C}_{18}\text{H}_{36}\text{O}_2(s)) - 26 \text{ mol } (\Delta H_f^{\circ}, \text{O}_2(g)) \\ = 18 \text{ mol } (-393.5 \text{ kJ/mol}) + 18 \text{ mol } (-241.826 \text{ kJ/mol}) - 1 \text{ mol } (-948 \text{ kJ/mol}) - 26 \text{ mol } (0 \text{ kJ/mol}) \\ = \mathbf{-10,488 \text{ kJ}}$$

$$\text{c) } q(\text{kJ}) = (1.00 \text{ g C}_{18}\text{H}_{36}\text{O}_2) \left(\frac{1 \text{ mol C}_{18}\text{H}_{36}\text{O}_2}{284.47 \text{ g C}_{18}\text{H}_{36}\text{O}_2} \right) \left(\frac{-10,488 \text{ kJ}}{1 \text{ mol C}_{18}\text{H}_{36}\text{O}_2} \right) \\ = \mathbf{-36.9 \text{ kJ}}$$

$$q(\text{kcal}) = (-36.9 \text{ kJ}) \left(\frac{1 \text{ kcal}}{4.184 \text{ kJ}} \right) = \mathbf{-8.81 \text{ kcal}}$$

$$\text{d) } q(\text{kcal}) = 8.81 \text{ kcal/g} \times 11.0 \text{ g} = \mathbf{96.9 \text{ kcal}}$$

The calculated calorie content is consistent with the package information.

- 6.105 a) The heat of reaction is calculated from the heats of formation found in Appendix B. The ΔH_f° 's for all of the species, except SiCl_4 , are found in Appendix B. Use reaction 3, with its given $\Delta H_{\text{rxn}}^\circ$, to find $\Delta H_f^\circ [\text{SiCl}_4(\text{g})]$.

$$\Delta H_{\text{rxn}}^\circ = \Delta H_f^\circ [\text{SiO}_2(\text{s})] + 4 \Delta H_f^\circ [\text{HCl}(\text{g})] - [\Delta H_f^\circ [\text{SiCl}_4(\text{g})] + 2 \Delta H_f^\circ [\text{H}_2\text{O}(\text{g})]]$$
$$-139.5 \text{ kJ} = (-910.9 \text{ kJ}) + 4(-92.31 \text{ kJ}) - \Delta H_f^\circ [\text{SiCl}_4(\text{g})] - 2(-241.826 \text{ kJ})$$

$$\Delta H_f^\circ [\text{SiCl}_4(\text{g})] = -656.988 \text{ kJ/mol (unrounded)}$$

The heats of reaction for the first two steps can now be calculated.

$$\Delta H_{\text{rxn1}}^\circ = \Delta H_f^\circ [\text{SiCl}_4(\text{g})] - \Delta H_f^\circ [\text{Si}(\text{s})] + 2 \Delta H_f^\circ [\text{Cl}_2(\text{g})]$$

$$\Delta H_{\text{rxn1}}^\circ = -656.988 \text{ kJ} - 0 - 2(0) = -656.988 = \mathbf{-657.0 \text{ kJ}}$$

$$\Delta H_{\text{rxn2}}^\circ = \Delta H_f^\circ [\text{SiCl}_4(\text{g})] + 2 \Delta H_f^\circ [\text{CO}(\text{g})] - \Delta H_f^\circ [\text{SiO}_2(\text{g})] - 2 \Delta H_f^\circ [\text{C}(\text{gr})] - 2 \Delta H_f^\circ [\text{Cl}_2(\text{g})]$$

$$\Delta H_{\text{rxn2}}^\circ = -656.988 \text{ kJ} + 2(-110.5 \text{ kJ}) - (-910.9 \text{ kJ}) - 2(0) - 2(0) = 32.912 = \mathbf{32.9 \text{ kJ}}$$

b) Adding reactions 2 and 3 yield: $2 \text{ C}(\text{gr}) + 2 \text{ Cl}_2(\text{g}) + 2 \text{ H}_2\text{O}(\text{g}) \rightarrow 2 \text{ CO}(\text{g}) + 4 \text{ HCl}(\text{g})$

$$\Delta H_{\text{rxn2+3}}^\circ = 32.912 \text{ kJ} + (-139.5 \text{ kJ}) = -106.588 = \mathbf{-106.6 \text{ kJ}}$$

Confirm this result by calculating $\Delta H_{\text{rxn}}^\circ$ using Appendix B values.

$$\Delta H_{\text{rxn2+3}}^\circ = 2 \Delta H_f^\circ [\text{CO}(\text{g})] + 4 \Delta H_f^\circ [\text{HCl}(\text{g})] - 2 \Delta H_f^\circ [\text{C}(\text{gr})] - 2 \Delta H_f^\circ [\text{Cl}_2(\text{g})] - 2 \Delta H_f^\circ [\text{H}_2\text{O}(\text{g})]$$

$$\Delta H_{\text{rxn2+3}}^\circ = 2(-110.5 \text{ kJ}) + 4(-92.31 \text{ kJ}) - 2(0) - 2(0) - 2(-241.826 \text{ kJ}) = -106.588 = \mathbf{-106.6 \text{ kJ}}$$